

School of Computing and Information Systems
The University of Melbourne
COMP90049, Introduction to Machine Learning, Semester 1, 2026

Assignment 2 - Group Contract

Contract due: Monday, May 4 2026, 5 PM

Group Name:

Group Members:

- Zhenyu Zhang – 1255206 - zhenyuzhang2@student.unimelb.edu.au
- Xinkai Zhang – 1619423 - xinkaiz@student.unimelb.edu.au
- Yiming Zhang – 1681020 - yzhang4211@student.unimelb.edu.au
- Yifan Wu – 1618332 - yifanwu@student.unimelb.edu.au

Research Question:

This project aims to investigate the effectiveness of machine learning methods for predicting a continuous target variable (e.g., housing prices) based on structured tabular data.

The main objectives are to understand how different modelling choices and data characteristics influence predictive performance, as well as to analyse the role of features and preprocessing techniques in improving model generalisation.

The specific research questions include:

1. To what extent can machine learning models accurately predict the target variable based on available features?
2. How do different types of models (e.g., linear models, instance-based methods, and neural networks) compare in terms of predictive performance and generalisation ability?
3. How do different data preprocessing techniques (e.g., feature scaling, handling missing values) affect model performance?
4. Which features contribute most significantly to the prediction task, and how can feature importance be interpreted?

Project Overview:

We will follow a standard machine learning pipeline to complete this project.

First, we will select a publicly available structured dataset with sufficient size and complexity. The dataset will be preprocessed by handling missing values, encoding categorical variables, and applying feature scaling where necessary.

Next, we will implement multiple machine learning models, including linear models, instance-based methods, and neural networks, to perform the prediction task.

We will evaluate model performance using appropriate metrics (e.g., RMSE, R^2) and apply cross-validation to ensure reliable results.

In addition, we will investigate the impact of different preprocessing techniques and conduct feature importance analysis to better understand the underlying data patterns.

Finally, we will compare results across models and settings, and provide a detailed discussion of findings, limitations, and potential improvements.

Roles and Responsibilities:

Specify the roles each group member will undertake throughout the project. Define the tasks, responsibilities, and contributions expected from each member. Ensure that roles are distributed fairly and leverage each member's strengths.

Member Name	Role and Responsibilities
Yifan Wu	Responsible for data preprocessing, including data cleaning, handling missing values, encoding categorical variables, and basic feature engineering. Also contributes to exploratory data analysis and report writing.
Xinkai Zhang	Responsible for implementing baseline and traditional machine learning models (e.g., Linear Regression, KNN), including training, tuning, and initial performance analysis. Contributes to experimental design and report writing.
Zhenyu Zhang	Responsible for model evaluation and optimisation, including hyperparameter tuning and applying evaluation metrics (e.g., RMSE, R^2 , cross-validation). Also supports neural network implementation and contributes to analysis and report writing.
Yiming Zhang	Responsible for results integration and analysis, including comparing model performance, interpreting findings, and coordinating report structure and final editing. Also contributes to selected model implementation if needed.

Communication Plan:

The group will primarily communicate via WeChat/Teams for daily discussions and quick updates.

Important decisions, experimental results, and progress updates will be documented in a shared GitHub repository and/or Google Docs.

All members are expected to respond to messages within 12 hours. For urgent issues (e.g., approaching deadlines), responses should be provided within 3 hours where possible.

Meeting Schedule:

The group will meet 2 times per week, with each meeting lasting approximately 1 hour each.

Meetings will focus on progress updates, problem-solving, and planning next steps. Additional meetings may be scheduled before key deadlines.

All members are expected to attend meetings on time and actively participate. If a member cannot attend, prior notice must be given.

Decision-Making Process:

Decisions will be made through group discussion, with the goal of reaching consensus.

If consensus cannot be achieved, a majority vote will be used. For technical disagreements, decisions should be supported by experimental results or references where possible.

In cases of uncertainty, simpler and more feasible approaches will be prioritised to ensure timely progress.

Work Plan and Timeline:

May 3 – May 5:

- Finalise dataset selection and research questions
- Set up working environment and data pipeline

May 6 – May 9:

- Perform data preprocessing (missing values, encoding, scaling)
- Conduct exploratory data analysis (EDA)
- Implement baseline models (e.g., Linear Regression, KNN)

May 10 – May 14:

- Implement additional models (e.g., SVM, Neural Network)
- Perform model comparison using cross-validation
- Begin preprocessing experiments (e.g., scaling, missing value strategies)

May 15 – May 18:

- Conduct feature importance analysis
- Finalise model evaluation and interpret results
- Identify key findings and limitations

May 19 – May 21:

- Complete report writing (all sections)
- Internal review and revision
- Final formatting and consistency checks

May 22:

- Final submission

Code of Conduct:

All members are expected to communicate respectfully and consider different perspectives during discussions.

Each member is responsible for completing assigned tasks on time and contributing to both technical work and report writing.

If a member is unable to complete their tasks, they must inform the group as early as possible.

Repeated lack of contribution without valid reason may be reflected in peer evaluation.

Disagreements or non-responsiveness:

In the event of disagreements or non-responsiveness, the group will follow a three-step process:

1. Communication Attempt:

The group will first attempt to resolve the issue through direct communication with the concerned member.

2. Internal Discussion:

If the issue persists, it will be discussed collectively within the group to reach a resolution.

3. Escalation:

If no agreement can be reached, the issue will be escalated to the tutor for mediation, and if necessary, further to the subject coordinator.

All members are expected to engage in this process in good faith and prioritise the success of the group.

Signature:

By signing below, each group member acknowledges their commitment to adhere to the terms outlined in this contract.

Student 1: _____ Date: _____

Student 2: _____ Date: _____

Student 3: _____ Date: _____

Student 4: _____ Date: _____

This contract is intended to establish clear expectations and promote effective collaboration among group members throughout the duration of the project. Any amendments to this contract should be discussed and agreed upon by all group members.